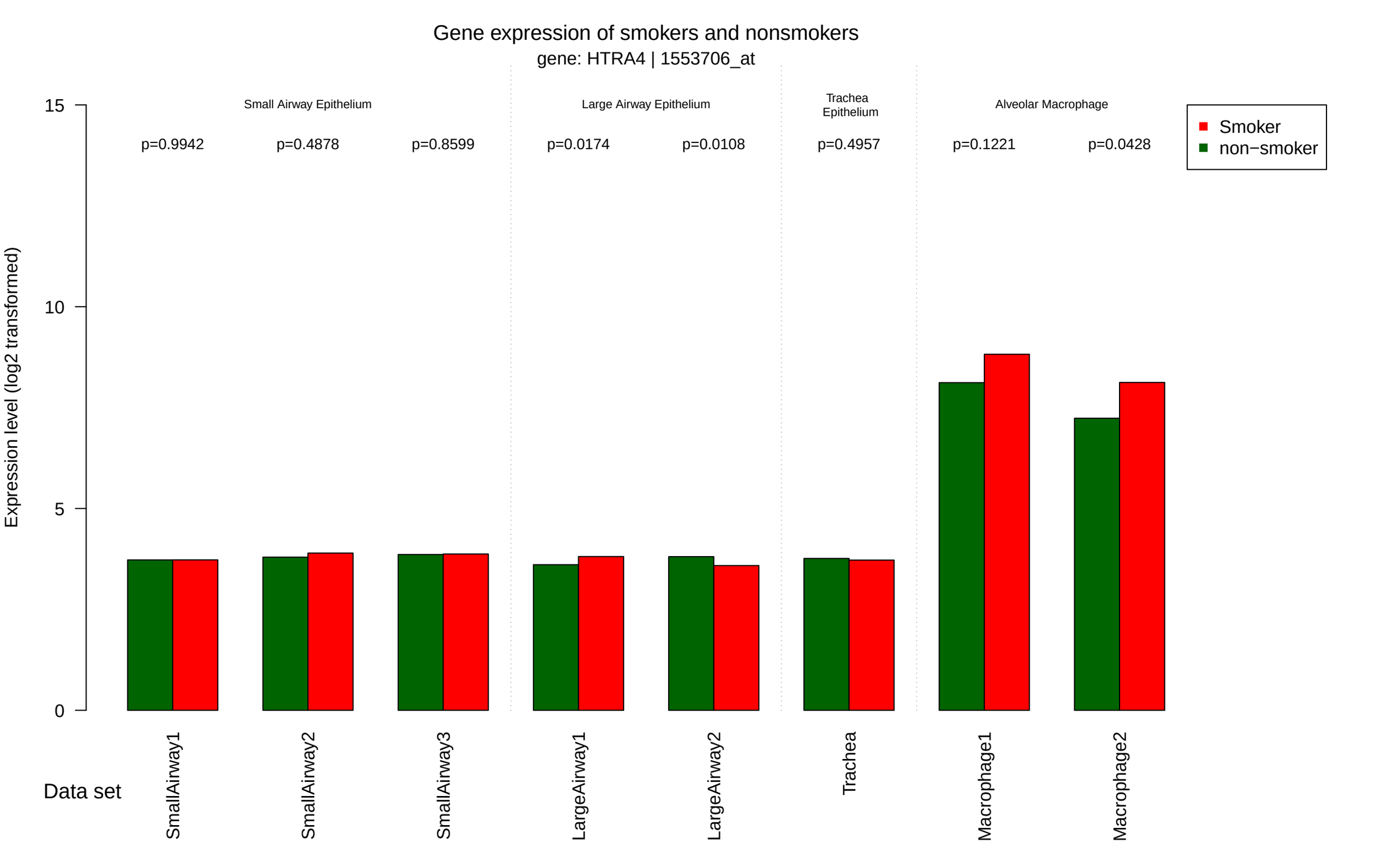
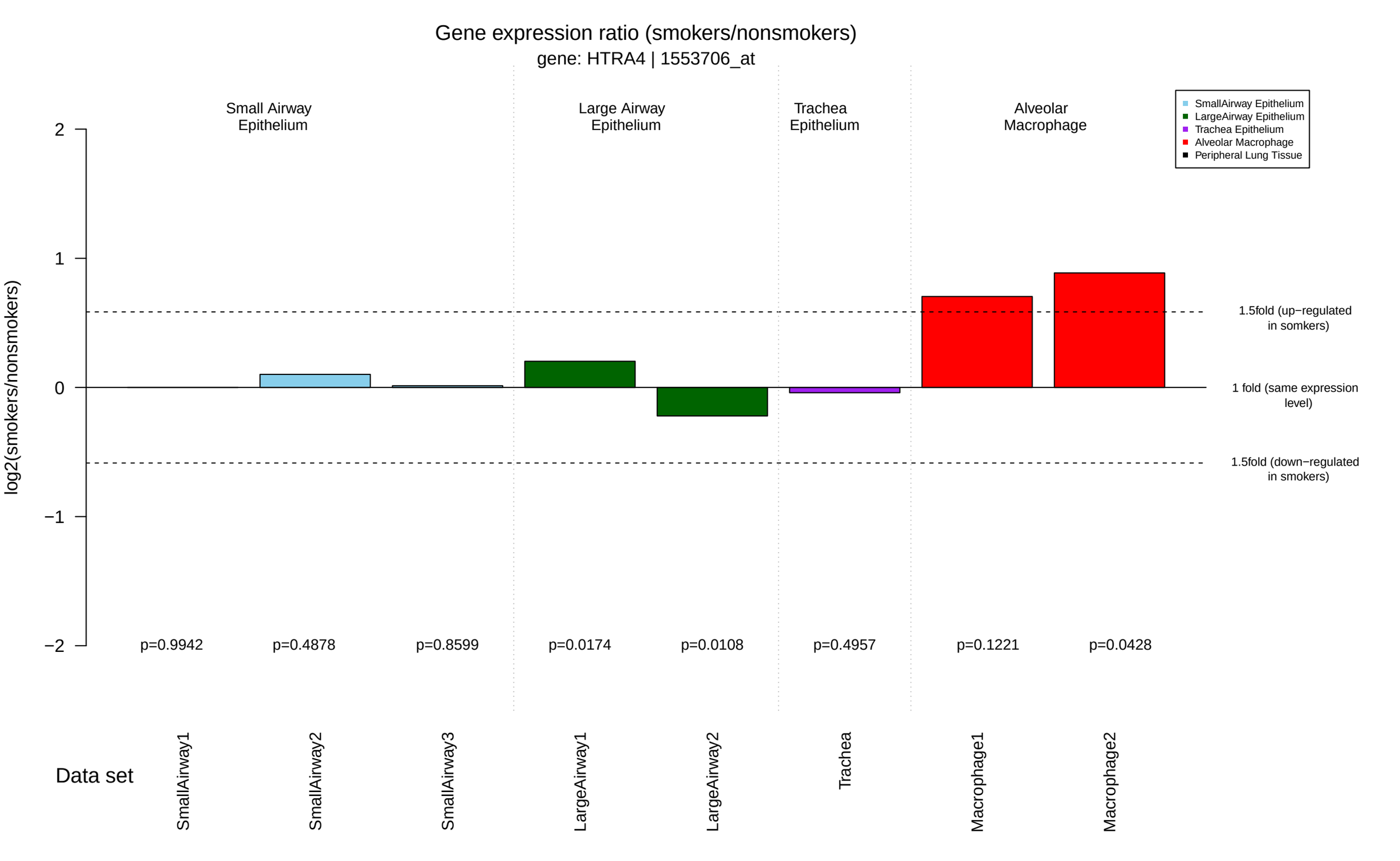
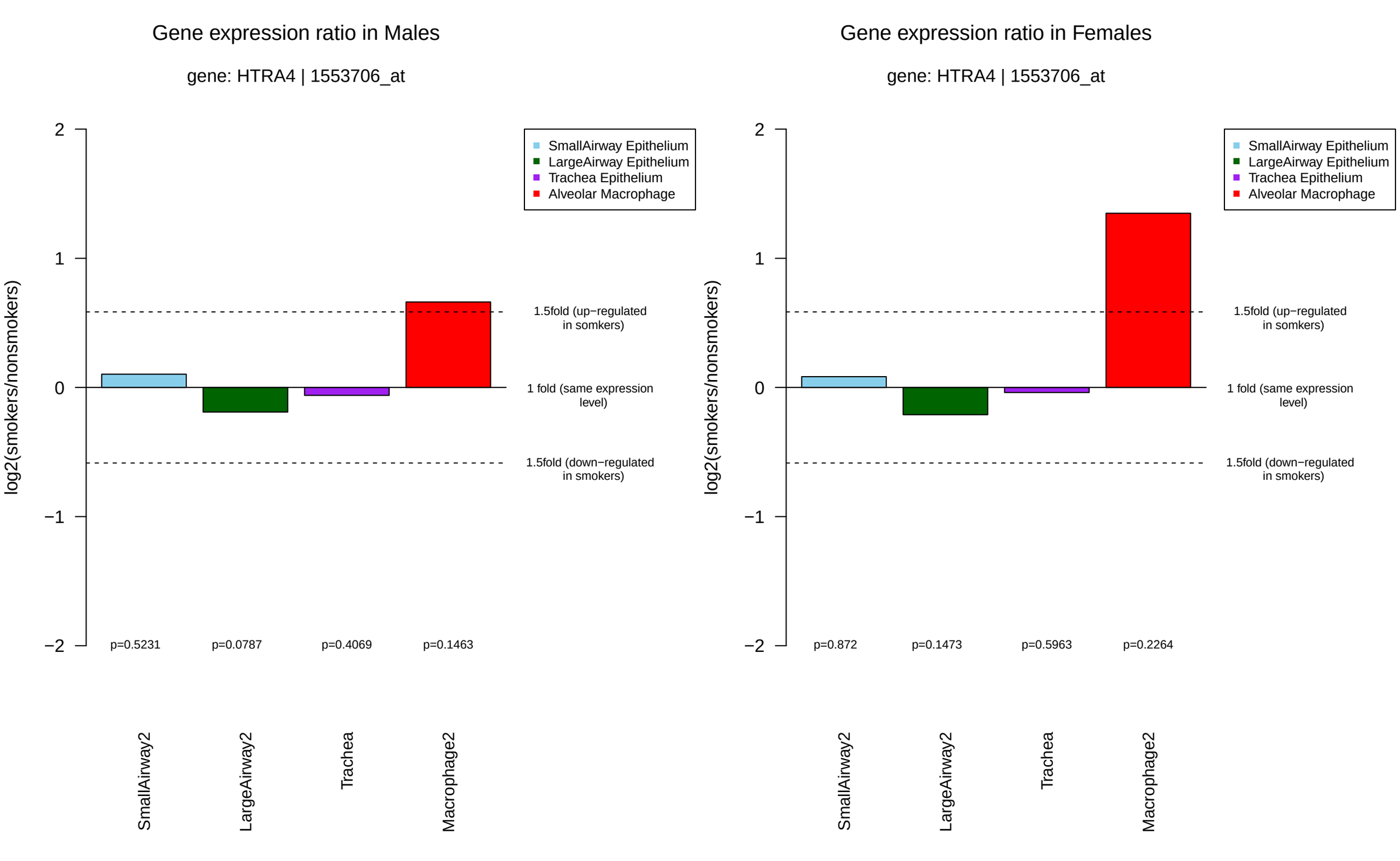
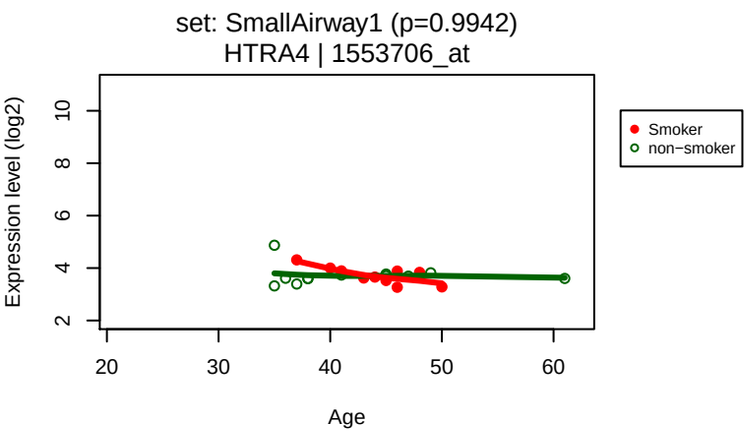
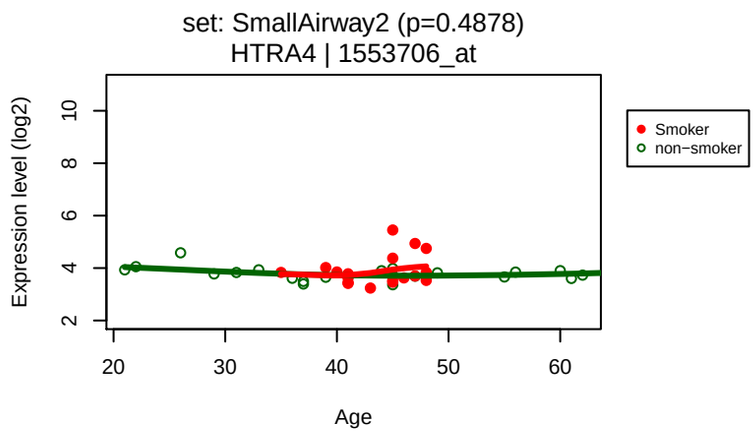
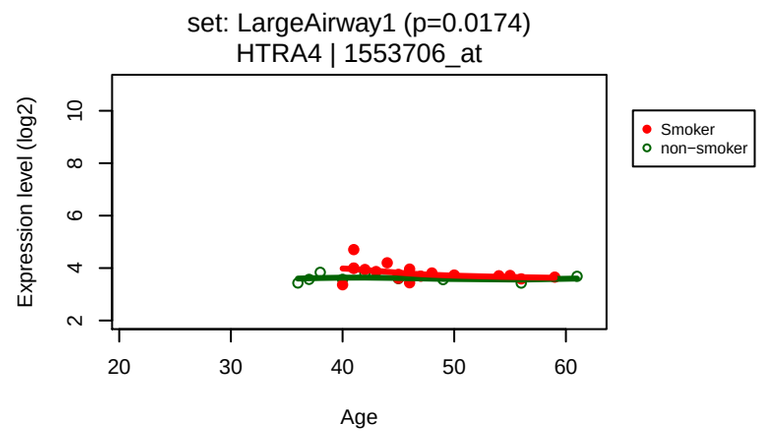

Gene expression of smokers and nonsmokers
gene: HTRA4 | 1553706_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9942
p=0.4878
p=0.8599
p=0.0174
p=0.0108
p=0.4957
p=0.1221
p=0.0428
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

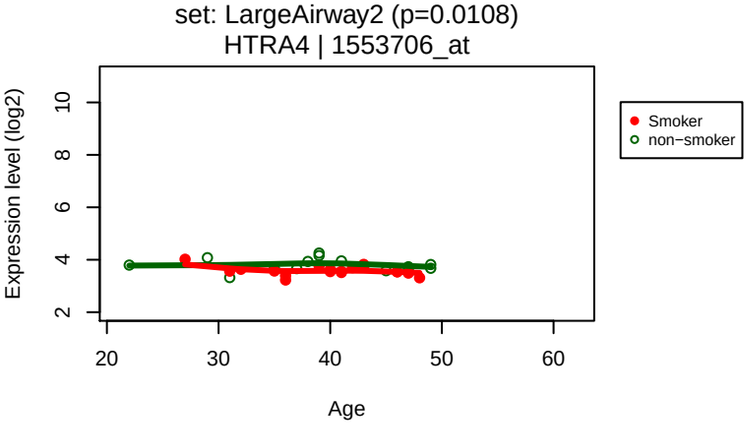

Gene expression ratio (smokers/nonsmokers)
gene: HTRA4 | 1553706_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9942
p=0.4878
p=0.8599
p=0.0174
p=0.0108
p=0.4957
p=0.1221
p=0.0428
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

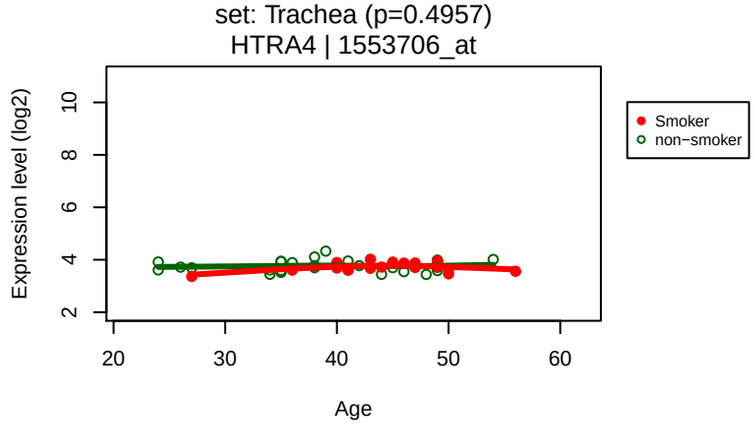

Gene expression ratio in Males
gene: HTRA4 | 1553706_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5231
p=0.0787
p=0.4069
p=0.1463
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HTRA4 | 1553706_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.872
p=0.1473
p=0.5963
p=0.2264
SmallAirway2
LargeAirway2
Trachea
Macrophage2

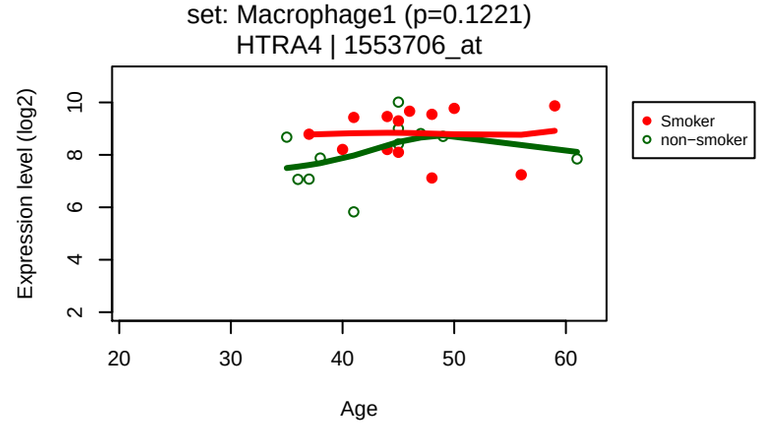

set: SmallAirway1 (p=0.9942)
HTRA4 | 1553706_at
Expression level (log2)
Age
Smoker
non−smoker

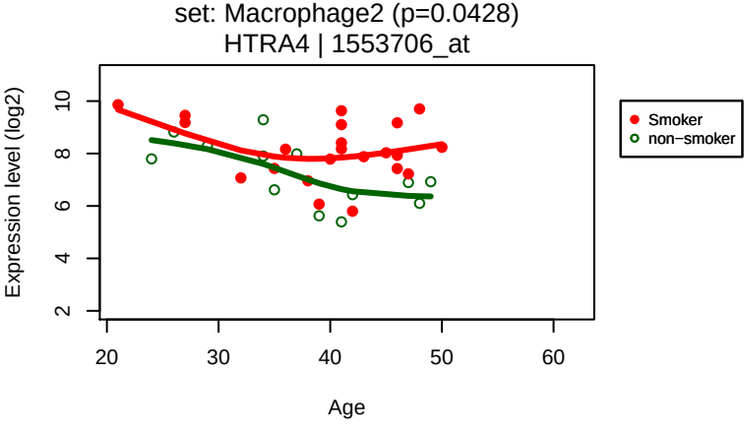

set: SmallAirway2 (p=0.4878)
HTRA4 | 1553706_at
Expression level (log2)
Age
Smoker
non−smoker

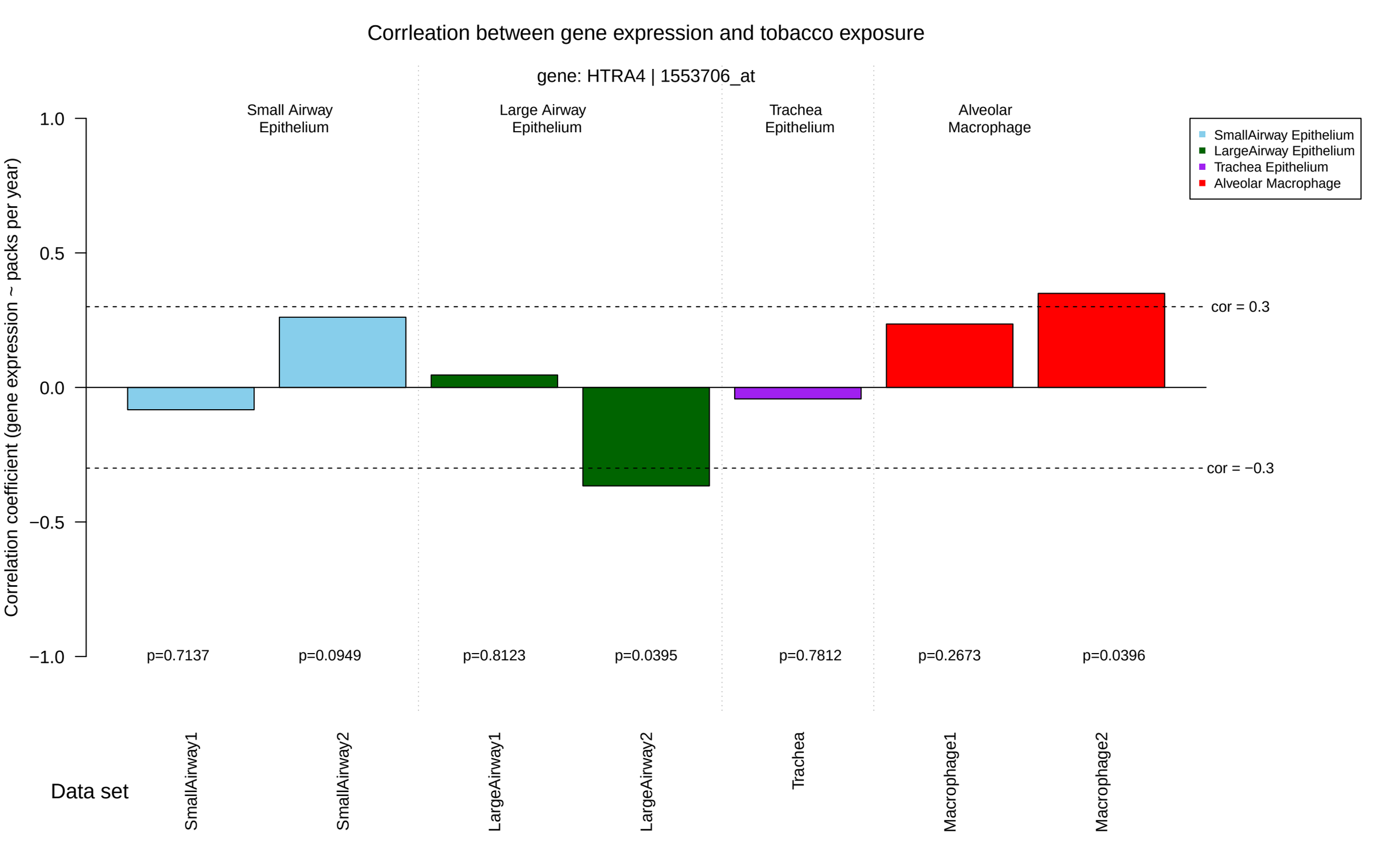

set: LargeAirway1 (p=0.0174)
HTRA4 | 1553706_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0108)
HTRA4 | 1553706_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4957)
HTRA4 | 1553706_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1221)
HTRA4 | 1553706_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0428)
HTRA4 | 1553706_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HTRA4 | 1553706_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7137
p=0.0949
p=0.8123
p=0.0395
p=0.7812
p=0.2673
p=0.0396
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2